ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

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Project Name _____
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17.12IO Structural Checklist for Building Types C2: Concrete Shear Walls with Stiff Diaphragms and C2a: Concrete Shear Walls with Flexible Diaphragms

Table 17-25. Immediate Occupancy Structural Checklist for Building Types C2 and C2a

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE BEARING WALLS: Floor and roof girders and trusses are not supported at the ends of concrete walls that are less than 10 in. (254 mm) thick. This statement only applies to framing supports located less than two times the wall thickness away from the wall end.	5.5.2.5.1	A.3.1.6.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the concrete shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f_c'}$.	5.5.3.1.1	A.3.2.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REINFORCING STEEL: The ratio of reinforcing steel area to gross concrete area is not less than 0.0012 in the vertical direction and 0.0020 in the horizontal direction. The spacing of reinforcing steel is equal to or less than 18 in. (457.2 mm).	5.5.3.1.3	A.3.2.2.2	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WALL ANCHORAGE AT FLEXIBLE DIAPHRAGMS: Exterior concrete or masonry walls that are dependent on flexible diaphragms for lateral support are anchored for out-of-plane forces at each diaphragm level with steel anchors, reinforcing dowels, or straps that are developed into the diaphragm. Connections have strength to resist the connection force calculated in the Quick Check procedure of Section 4.4.3.7.	5.7.1.1	A.5.1.1	

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C	NC	N/A	U			
☐	☐	☐	☐	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of loads to the shear walls, and the connections are able to develop the lesser of the shear strength of the walls or diaphragms.	5.7.2	A.5.2.1
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	FOUNDATION DOWELS: Wall reinforcement is doweled into the foundation, and the dowels are able to develop the lesser of the strength of the walls or the uplift capacity of the foundation.	5.7.3.4	A.5.3.5

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Seismic-Force-Resisting System							
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	DEFLECTION COMPATIBILITY: Secondary components have the shear capacity to develop the flexural strength of the components and are compliant with the following items in Table 17-23: COLUMN-BAR SPLICES, BEAM-BAR SPLICES, COLUMN-TIE SPACING, STIRRUP SPACING, and STIRRUP AND TIE HOOKS.	5.5.2.5.2	A.3.1.6.2	
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	FLAT SLABS: Flat slabs or plates not part of seismic-force-resisting system have continuous bottom steel through the column joints.	5.5.2.5.3	A.3.1.6.3	
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	COUPLING BEAMS: Coupling beams have the capacity in shear to develop the uplift capacity of the adjacent wall or to develop the flexural capacity of the coupling beam, whichever is less. This statement only applies to coupling beams with span-to-depth ratios exceeding 2-to-1.	5.5.3.2.1	A.3.2.2.3	
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	OVERTURNING: All shear walls have aspect ratios less than 4-to-1. Wall piers need not be considered.	5.5.3.1.4	A.3.2.2.4	
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	CONFINEMENT REINFORCING: For shear walls with aspect ratios greater than 2-to-1, the boundary elements are confined with spirals or ties with spacing less than $8d_b$.	5.5.3.2.2	A.3.2.2.5	

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C	NC	N/A	U			
☐	☐	☐	☐	WALL REINFORCING AT OPENINGS: There is added trim reinforcement around all wall openings with a dimension greater than three times the thickness of the wall.	5.5.3.1.5	A.3.2.2.6
☐	☐	☐	☐	WALL THICKNESS: Thicknesses of bearing walls are not less than 1/25 the unsupported height or length, whichever is shorter, nor less than 4 in. (101.6 mm).	5.5.3.1.2	A.3.2.2.7
Diaphragms (Stiff or Flexible)						
☐	☐	☐	☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1
☐	☐	☐	☐	ROOF CHORD CONTINUITY: All chord elements are continuous, regardless of changes in roof elevation.	5.6.1.1	A.4.1.3
☐	☐	☐	☐	OPENINGS AT SHEAR WALLS: Diaphragm openings immediately adjacent to the shear walls are less than 15% of the wall length.	5.6.1.3	A.4.1.4
☐	☐	☐	☐	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7
☐	☐	☐	☐	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8
Flexible Diaphragms						
☐	☐	☐	☐	CROSSTIES: There are continuous crossties between diaphragm chords.	5.6.1.2	A.4.1.2
☐	☐	☐	☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 12 ft (3.6 m) and aspect ratios less than 1-to-1 in the direction being considered.	5.6.2	A.4.2.1
☐	☐	☐	☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 30 ft (9.2 m) and aspect ratios less than or equal to 3-to-1.	5.6.2	A.4.2.2

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C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BLOCKED DIAPHRAGMS: All blocked wood structural panel diaphragms have horizontal spans less than 90 ft (27.4 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.3
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CANTILEVERED WOOD DIAPHRAGMS: All cantilevered wood diaphragms that provide lateral support for concrete or masonry walls consist of wood structural panels and have a maximum cantilever length of 15 ft (4.6 m) if unblocked or 25 ft (7.6 m) if blocked, and a maximum ratio of cantilever length to diaphragm width of 1:2.5 if unblocked and 1:1.5 if blocked. In addition, the cantilevered diaphragm has a back-span length equal to or greater than the cantilevered portion.	5.6.2	A.4.2.4
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced concrete consist of horizontal spans of less than 40 ft (12.2 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1
Connections						
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps; the pile cap reinforcement and pile anchorage are able to develop the tensile capacity of the piles.	5.7.3.5	A.5.3.8

Status				**Evaluation Statement**	**Tier 2 Reference**	**Commentary Reference**	**Comments**
Moderate and High Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Foundation System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1	

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